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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the triangulated-subcategory framework, the distinguished triangle comparing the powers of I, the rewrite of the associated graded piece, and the induction by saturation using the tensor product lemma.

Editorial note. The label lemma-locally-constant-bounded names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Locally constant finite type modules form a weak Serre subcategory. Modules on Sites, the lemma describing the exactness properties of locally constant sheaves of finite type modules. Supplies that the complexes with such cohomology sheaves form a saturated triangulated subcategory.
- The triangulated subcategory criterion. Derived Categories, the lemma describing the full subcategory of complexes whose cohomology lies in a Serre subcategory. Supplies the framework in which the induction over the powers of I is carried out.
- Tensor products of locally constant finite type sheaves. Same chapter, the lemma showing that the derived tensor product of a locally constant finite type module with another such module stays locally constant of finite type. Supplies the final step identifying the associated graded pieces.

Source reference. Original lemma 06UQ: sources/stacks-chow/tags.txt, source lines 8754–8754. Prerequisite pointer invoked at proof line(s) 14399; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 094H: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 14356–14356. Prerequisite pointer invoked at proof

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line(s) 14414; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 093U: sources/stacks-chow/tags.txt, source lines 11527–11527. Prerequisite pointer invoked at proof line(s) 14394; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let $K \in D^-(\mathcal{C}, \Lambda)$. If the cohomology sheaves of $K \otimes^{\mathbf{L}}_{\Lambda} \underline{\Lambda/I}$ are locally constant sheaves of Λ/I-modules of finite type, then the cohomology sheaves of $K \otimes^{\mathbf{L}}_{\Lambda} \underline{\Lambda/I^n}$ are locally constant sheaves of Λ/I^n-modules of finite type for all $n \geq 1$.*

Proof. Recall that the locally constant sheaves of Λ-modules of finite type form a weak Serre subcategory of all $\underline{\Lambda}$-modules, see Modules on Sites, Lemma 093U. Thus the subcategory of $D(\mathcal{C}, \Lambda)$ consisting of complexes whose cohomology sheaves are locally constant sheaves of Λ-modules of finite type forms a strictly full, saturated triangulated subcategory of $D(\mathcal{C}, \Lambda)$, see Derived Categories, Lemma 06UQ. Next, consider the distinguished triangles

$$K \otimes^{\mathbf{L}}_{\Lambda} \underline{I^n/I^{n+1}} \to K \otimes^{\mathbf{L}}_{\Lambda} \underline{\Lambda/I^{n+1}} \to K \otimes^{\mathbf{L}}_{\Lambda} \underline{\Lambda/I^n} \to K \otimes^{\mathbf{L}}_{\Lambda} \underline{I^n/I^{n+1}}[1]$$

and the isomorphisms

$$K \otimes^{\mathbf{L}}_{\Lambda} \underline{I^n/I^{n+1}} = \left(K \otimes^{\mathbf{L}}_{\Lambda} \underline{\Lambda/I}\right) \otimes^{\mathbf{L}}_{\Lambda/I} \underline{I^n/I^{n+1}}$$

Combined with Lemma 094H we obtain the result. □